MACHINE LEARNING FUNDAMENTALS - LS2026
SEMINAR: GENERATIVE LEARNING

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Assignment 1. Consider a multiclass classification problem with the following setup. The input space is $\mathcal{X} = \mathbb{R}$, the label space is $\mathcal{Y} = \{1, \ldots, Y\}$, and the loss function is $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_+$. You are given an i.i.d. training sample

$$T_m = \big((x_i, y_i) \in \mathcal{X} \times \mathcal{Y} \mid i = 1, \ldots, m\big).$$

The task consists of the following parts:

a) Fit the training data using a mixture of univariate normal distributions,

$$p(x, y; \boldsymbol{\theta}) = \frac{p(y)}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu_y)^2}{2\sigma^2} \right).$$

The parameters $\boldsymbol{\theta}$ of the mixture model include: i) the prior probabilities $p(y)$, $y \in \mathcal{Y}$, ii) the class-conditional means $\mu_y \in \mathbb{R}$, $y \in \mathcal{Y}$, and iii) the variance $\sigma^2 \in \mathbb{R}_{++}$. Estimate all parameters of the mixture model using the maximum likelihood method.

b) Derive the formulas for the plug-in Bayes classifier. Then specialize the result to the $0/1$-loss function.

c) To quantify prediction uncertainty, derive a formula for the plug-in conditional risk. Then instantiate the conditional risk for the $0/1$-loss function.

Solution 1. For each class $y \in \mathcal{Y}$, define

$$\mathcal{I}_y := \{i \in \{1, \ldots, m\} \mid y_i = y\}, \qquad n_y := |\mathcal{I}_y|.$$

Then $\sum_{y=1}^{Y} n_y = m$.

A) MAXIMUM LIKELIHOOD ESTIMATION

Since the training sample is i.i.d., the log-likelihood is

$$
\begin{aligned}
L(T_m, \boldsymbol{\theta}) &= \log \left(\prod_{i=1}^{m} p(x_i, y_i; \boldsymbol{\theta}) \right) \\
&= \log \left(\prod_{i=1}^{m} p(y_i) \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_i - \mu_{y_i})^2}{2\sigma^2} \right) \right) \\
&= \sum_{i=1}^{m} \log p(y_i) - m \log(\sqrt{2\pi}\sigma) - \frac{1}{2\sigma^2} \sum_{i=1}^{m} (x_i - \mu_{y_i})^2 \\
&= \sum_{i=1}^{m} \log p(y_i) - m \log(\sqrt{2\pi}\sigma) - \frac{1}{2\sigma^2} \sum_{y=1}^{Y} \sum_{i \in \mathcal{I}_y} (x_i - \mu_{y_i})^2
\end{aligned}
$$

Estimation of the priors $p(y)$. Since class y occurs n_y times in the sample,

$$\sum_{i=1}^{m} \log p(y_i) = \sum_{y=1}^{Y} n_y \log p(y).$$

Therefore, we maximize

$$\sum_{y=1}^{Y} n_y \log p(y)$$

subject to the constraint

$$\sum_{y=1}^{Y} p(y) = 1.$$

Using a Lagrange multiplier λ, define

$$F(p, \lambda) = \sum_{y=1}^{Y} n_y \log p(y) + \lambda \left(\sum_{y=1}^{Y} p(y) - 1 \right).$$

Differentiating with respect to $p(y)$ gives

$$\frac{\partial F}{\partial p(y)} = \frac{n_y}{p(y)} + \lambda = 0,$$

hence

$$p(y) = -\frac{n_y}{\lambda}.$$

Summing over y and using $\sum_{y=1}^{Y} p(y) = 1$ and $\sum_{y=1}^{Y} n_y = m$, we obtain

$$-\frac{m}{\lambda} = 1, \qquad \text{so} \qquad \lambda = -m.$$

Thus

$$\boxed{\hat{p}(y) = \frac{n_y}{m}.}$$

Estimation of the class means μ_y. The log-likelihood term depending on μ_y is

$$-\frac{1}{2\sigma^2}\sum_{i=1}^{m}(x_i-\mu_{y_i})^2 = -\frac{1}{2\sigma^2}\sum_{y=1}^{Y}\sum_{i\in I_y}(x_i-\mu_y)^2.$$

Hence, for each fixed class y, we minimize

$$\sum_{i\in I_y}(x_i-\mu_y)^2.$$

Differentiating with respect to μ_y yields

$$\frac{\partial}{\partial\mu_y}\sum_{i\in I_y}(x_i-\mu_y)^2 = -2\sum_{i\in I_y}(x_i-\mu_y).$$

Setting this derivative equal to zero, we get

$$\sum_{i\in I_y}(x_i-\mu_y)=0 \qquad \Longrightarrow \qquad n_y\mu_y=\sum_{i\in I_y}x_i.$$

Therefore,

$$\boxed{\hat{\mu}_y=\frac{1}{n_y}\sum_{i\in I_y}x_i\ .}$$

Estimation of the common variance σ^2. Substituting the estimated means into the log-likelihood gives

$$L(T_m,\boldsymbol{\theta}) = \text{const} - \frac{m}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{m}(x_i-\hat{\mu}_{y_i})^2.$$

Differentiate with respect to σ^2:

$$\frac{\partial L}{\partial\sigma^2} = -\frac{m}{2\sigma^2}+\frac{1}{2(\sigma^2)^2}\sum_{i=1}^{m}(x_i-\hat{\mu}_{y_i})^2.$$

Setting this derivative to zero gives

$$-\frac{m}{2\sigma^2}+\frac{1}{2(\sigma^2)^2}\sum_{i=1}^{m}(x_i-\hat{\mu}_{y_i})^2=0.$$

Multiplying by $2(\sigma^2)^2$, we obtain

$$-m\sigma^2+\sum_{i=1}^{m}(x_i-\hat{\mu}_{y_i})^2=0,$$

and hence

$$\boxed{\hat{\sigma}^2=\frac{1}{m}\sum_{i=1}^{m}(x_i-\hat{\mu}_i)^2\ .}$$

B) PLUG-IN BAYES CLASSIFIER

The Bayes classifier reads:

$$h_*(x) = \arg\min_{\hat{y}\in\mathcal{Y}} \sum_{y\in\mathcal{Y}} p(y \mid x)\ell(\hat{y}, y) .$$

In the plug-in approach, the unknown posterior probabilities are replaced by their estimates under the fitted model. Thus the plug-in Bayes classifier is

$$\boxed{\hat{h}(x) = \arg\min_{\hat{y}\in\mathcal{Y}} \sum_{y\in\mathcal{Y}} \hat{p}(y \mid x)\ell(y, \hat{y})} .$$

Posterior probabilities. By Bayes' rule,

$$\hat{p}(y \mid x) = \frac{\hat{p}(y)\,\dfrac{1}{\sqrt{2\pi}\hat{\sigma}}\exp\left(-\dfrac{(x-\hat{\mu}_y)^2}{2\hat{\sigma}^2}\right)}{\sum_{k=1}^{Y}\hat{p}(k)\,\dfrac{1}{\sqrt{2\pi}\hat{\sigma}}\exp\left(-\dfrac{(x-\hat{\mu}_k)^2}{2\hat{\sigma}^2}\right)} .$$

The factor $\frac{1}{\sqrt{2\pi}\hat{\sigma}}$ is common to all classes and cancels out, hence

$$\boxed{\hat{p}(y \mid x) = \frac{\hat{p}(y)\exp\left(-\dfrac{(x-\hat{\mu}_y)^2}{2\hat{\sigma}^2}\right)}{\sum_{k=1}^{Y}\hat{p}(k)\exp\left(-\dfrac{(x-\hat{\mu}_k)^2}{2\hat{\sigma}^2}\right)}} .$$

Specialization to the $0/1$-loss. For the $0/1$-loss $\ell(y, y') = [\![y \neq y']\!]$, the plugin Bayes classifier simplifies to:

$$\begin{aligned}\hat{h}(x) &= \arg\max_{y\in\mathcal{Y}} \hat{p}(y \mid x) \\ &= \arg\max_{y\in\mathcal{Y}} \log\left(\hat{p}(y)\,\frac{1}{\sqrt{2\pi}\hat{\sigma}}\exp\left(-\frac{(x-\hat{\mu}_y)^2}{2\hat{\sigma}^2}\right)\right) \\ &= \arg\max_{\hat{y}\in\mathcal{Y}} \left(\log\hat{p}(y) - \frac{(x-\hat{\mu}_y)^2}{2\hat{\sigma}^2}\right)\end{aligned}$$

C) PLUG-IN CONDITIONAL RISK AND PREDICTION UNCERTAINTY

The conditional risk associated with the prediction $h_*(x)$ reads:

$$r_*(x) = \sum_{y\in\mathcal{Y}} p(y \mid x)\ell(h_*(x), y) = \min_{\hat{y}\in\mathcal{Y}} \sum_{y\in\mathcal{Y}} p(y \mid x)\ell(\hat{y}, y) .$$

and the plug-in conditinal risk is

$$\hat{r}(x) = \min_{\hat{y}\in\mathcal{Y}} \sum_{y\in\mathcal{Y}} \hat{p}(y \mid x)\ell(\hat{y}, y) .$$

Specialization to the $0/1$-loss. For the $0/1$-loss,

$$\hat{r}(x) = \min_{\hat{y} \in \mathcal{Y}} \sum_{y \in \mathcal{Y}} \hat{p}(y \mid x) \ell(\hat{y}, y) = 1 - \max_{y \in \mathcal{Y}} \hat{p}(y \mid x) \ .$$